W H I T E P A P E R

Transient Supra-Coherence T-U° Operator

Theory, Formal Properties, and Governed Hyper-Coherence in Multi-Layer Invariant-Certified Autonomous Systems

Kemar Armando Morrison
Principal Investigator · Codex Sovereign Research
The Open University, London, UK

March 2026 · Codex Sovereign Research · Preprint — Not Peer Reviewed

Executive Summary

The Transient Supra-Coherence (TSC) operator, designated T-U°, is a formally derived, empirically validated measure of instantaneous multi-layer representational alignment in autonomous systems governed by the Absolute(0) Stability Framework (ASV3-FORMAL). This white paper presents the complete theoretical foundations, formal guarantees, and empirical evidence for the TSC operator and its associated hyper-coherence phenomenon.

At its core, TSC answers a single operational question at every governance step: how coherent are the system's internal representations across all layers, relative to its control inputs and future trajectory? A high TSC value certifies that layer states, the gradient correction kernel, and the future-adjacency vector are mutually aligned — a condition termed hyper-coherence — and that governance coupling should be correspondingly tightened.

Empirical validation across 80 benchmark runs (state-space dimensions $d \in \{64, 128, 256, 512, 1024\}$, four signal regimes, four defence domains, 500 steps each) confirms: zero invariant violations universally, 100% anchor lock across all conditions, and peak TSC values of 1.639 at $d = 1024$ under adversarial loading. Under adversarial conditions, the system spends 78.4% of operating time in the T-U° hyper-coherent regime — the highest-performance governance configuration.

Key Findings at a Glance

- Peak TSC Coherence Index: 1.639 ($d = 1024$, adversarial loading)
- Invariant Violations: Zero across all 80 benchmark runs
- Anchor Lock: 100% maintained universally across all dimensions and scenarios
- AGSPL Lock: Achieved at all readiness signals ≥ 0.85; zero auto-tune iterations at $\sigma \geq 0.95$
- Hyper-Coherence Prevalence: 78.4% of steps under adversarial conditions ($d = 1024$)

1. Introduction and Motivation

Stability governance frameworks for autonomous systems must do more than prevent invariant violations — they must measure and exploit the transient coherence of multi-layer internal states to accelerate recovery and tighten control during periods of high self-consistency. Prior governance frameworks (QBN v7 and earlier) lacked a coherence measure capable of distinguishing two states with identical spectral deviation but very different internal representational alignment.

Consider two systems sitting on the boundary of the admissible set. System A has layer states that are mutually consistent — all pointing in similar directions in the representational manifold.

System B has layer states that are fragmented — high-norm but incoherent. Both systems exhibit $\delta = 0$, yet System A can be governed with much tighter coupling, as its internal representations will reinforce rather than cancel each other's contributions to the correction field.

The TSC T-U° operator fills this gap. It measures the geometric consistency of three signal components: the governed state layers, the gradient correction kernel (the ultra-kernel), and the future trajectory adjustment. The resulting coherence index drives the Adaptive Gain-Scheduled Phase Lock (AGSPL) coupling efficiency, translating representational alignment directly into governance tightness.

T	Transient — valid only for the duration of multi-layer invariant compliance.
U	Ultra — coherence has exceeded the ultra-kernel baseline through constructive superposition of layer norms.
°	Degrees — the onset angle of supra-alignment in the Taleken phase space, measured relative to the admissible manifold boundary.

2. Theoretical Foundations

2.1 The Taleken–Völundr Framework

The broader theoretical context for the TSC operator is the Taleken–Völundr framework, a spectral admissibility engine that maintains the system state within a spectrally regular manifold via event-triggered gradient correction. The framework is grounded in a spectral observable algebra computed from the governed state $x \in X \subseteq \mathbb{R}^d$.

The central object is the Taleken concentration metric $\Delta(t)$, defined as the deviation of the spectral kurtosis from its admissibility threshold. The system is spectrally admissible if and only if $\Delta(t) \leq 0$. When $\Delta(t) > 0$, the Taleken–Völundr Law triggers event-driven gradient correction, directing the system back toward the admissible manifold via the AGGL correction field.

Crucially, this correction field is the source of the ultra-kernel $u^\star$ that feeds into TSC. During active Taleken correction, $u^\star$ is non-zero and large, elevating TSC(t) and potentially driving the system into the hyper-coherent regime. TSC thus converts an active governance response into a positive coherence signal, creating a virtuous feedback loop between admissibility enforcement and representational alignment.

2.2 The TSC Composite Index

The TSC operator is defined over the state space $X \subseteq \mathbb{R}^d$, where d is the state-space dimension. All vectors are elements of $\mathbb{R}^d$ with the standard Euclidean norm.

Dimension-Normalised Layer Norm

For a layer-state vector $s \in \mathbb{R}^d$, the dimension-normalised norm is:

$$\lambda(s) = \|s\| / (\sqrt{d} + \varepsilon)$$

where $\varepsilon > 0$ is a small regularisation constant. The normalisation by $\sqrt{d}$ renders λ dimension-independent in expectation for unit-variance inputs, enabling meaningful coherence comparisons across architecture instantiations of different depths and widths.

TSC Functional

Given layer states $L = \{s_1, \ldots, s_n\} \subset \mathbb{R}^d$, ultra-kernel $u \in \mathbb{R}^d$, and future-adjacency vector $f \in \mathbb{R}^d$:

$$\Phi = \Sigma_i \lambda(s_i) \quad \text{[layer superposition norm]}$$

$$u\bigstar = \lambda(u) \quad \text{[ultra-kernel norm]}$$

$$s\tau = \lambda(f) \cdot \min(\tau f, 2) \quad \text{[future-adjacency norm]}$$

$$TSC(t) = (\Phi + u\bigstar + s\tau) / (n + 2)$$

The denominator (n + 2) normalises by the total number of signal components — n governance layers plus the ultra-kernel and future-adjacency — ensuring TSC(t) is a per-component average comparable across systems with different numbers of governance layers.

GREEN-Phase Coherence Amplification

When the governance phase is GREEN and the current session is valid, TSC(t) receives a 1.05× multiplicative boost. This reflects the empirical observation that GREEN-phase coherence is more predictive of sustained stability than coherence achieved during recovery from RED. A system maintaining continuous admissibility has demonstrated a higher quality of governance — the boost rewards this distinction, creating a measurable coherence premium for sustained nominal operation.

3. The Hyper-Coherence Phenomenon

Hyper-coherence is the central phenomenon of the TSC framework. It is the regime in which TSC(t) exceeds the threshold θ_h, triggering qualitative changes in governance coupling and marking the onset of the T-U° state.

3.1 Onset Condition

The system enters the T-U° supra-coherence state at time t_{sx} when:

$$TSC(t_{sx}) > \theta_h \quad \wedge \quad dTSC/dt|t=t_{sx} \geq 0$$

The onset condition requires both threshold exceedance AND rising coherence, ensuring the system is entering — not exiting — the high-coherence regime. Hyper-coherence is most prevalent under adversarial and combined signal conditions, because adversarial inputs produce sustained spectral violations ($\Delta > 0$), which keep the Taleken correction gate open and the ultra-kernel u large.

3.2 Derivative Law and Supra-Alignment Ceiling

The instantaneous time derivative of TSC(t) is governed by:

$$dTSC/dt = [\Sigma_i \ \partial\Phi_i/\partial t + \partial u\bigstar/\partial t + \partial s\tau/\partial t] \ / \ (n + 2)$$

When all layer derivatives simultaneously vanish (perfect phase-lock), the TSC derivative collapses to a residual driven only by future-adjacency drift. The system approaches the supra-alignment ceiling:

$$\lim_{allllen5n4 \to max} \ dTSC/dt \to 1.0^+$$

The notation 1.0^+ indicates that the derivative approaches unity from above — the marginal rate of coherence growth equals the normalisation rate, representing maximal sustainable alignment. This finite ceiling is a direct consequence of the dimension-normalised layer norm, preventing runaway coherence accumulation under adversarial loading.

3.3 Finite Epoch Duration

A critical safety property: hyper-coherent epochs are finite in duration. Once the Taleken gate closes ($\Delta \leq 0$), the ultra-kernel becomes zero and TSC(t) falls below θ_h in at most:

$$K_hyper = \lceil(\theta_h - TSC_floor) \cdot (n+2) \ / \ \delta_decay\rceil \text{ steps}$$

This finiteness is a governance safety property, not a performance limitation. An unbounded hyper-coherent epoch would indicate the system never exits the inadmissible region. Finite epochs confirm the governance architecture successfully resolves the spectral violations that drive them.

4. Session Validity and Recovery

The TSC operator maintains a coherence session tracking the history of stable governance steps within a contiguous run. Sessions provide temporal context: a given TSC value carries more weight if sustained continuously from a clean starting state than if it follows immediately upon recovery from a collapse.

Condition	Session State	Transition	TSC Effect

Phase ∈ {GREEN, BLUE}	VALID	Normal step	TSC accumulated normally
RED ∧ ¬stable (< 3 steps)	VALID (degraded)	Warning step	No GREEN boost applied
RED ∧ ¬stable (≥ 3 steps)	INVALID	Collapse trigger	TSC reset; peak archived
GREEN/BLUE (after INVALID)	VALID (new session)	Recovery	session_count++; TSC restarts

Under the forward invariance guarantee of ASV3-FORMAL, the RED phase cannot persist indefinitely. The AGGL gradient correction reduces spectral deviation at each RED step, so the system exits RED in at most $K = \lceil \delta\star / (\kappa \cdot dt \cdot \|\text{grad } U\|_\text{min}) \rceil$ steps. Any session collapse therefore terminates in finite time, and session validity is restored.

5. Integration with Adaptive Gain-Scheduled Phase Lock

The TSC output is consumed by the Adaptive Gain-Scheduled Phase Lock (AGSPL) operator, which determines whether the governance pipeline is locked — whether the current readiness signal is sufficient to authorise full governance execution at maximum coupling. TSC boosts coupling efficiency through two additive mechanisms:

$$CE = \sigma \cdot g\star \cdot w + \beta_TSC + \beta_h \cdot \mathbf{1}[TSC(t) > \theta_h]$$

where $\beta_TSC = \alpha_TSC \cdot TSC(t)$ is the proportional TSC boost, and β_h is the discrete hyper-coherence bonus. The system locks (AGSPL_locked = True) if and only if $CE \geq \theta_sync$.

This design has two important consequences. First, during high-coherence episodes, lower readiness signals are sufficient to achieve lock — the system self-reinforces its own governance. Second, during hyper-coherent epochs, the discrete bonus β_h may achieve lock immediately, as confirmed empirically below.

Table 2: AGSPL Phase-Lock Signal Sweep (All Operational Modes)

Readiness (σ)	Coupling Eff.	Auto-Tune w	Iterations	Lock Status
0.85	0.5923	1.115	23	LOCKED
0.90	0.5934	1.055	11	LOCKED
0.95	0.5938	1.000	0	LOCKED
1.00	0.6250	1.000	0	LOCKED
1.05	0.6562	1.000	0	LOCKED

At readiness signal σ = 0.85, 23 auto-tune iterations are required to achieve lock. At σ ≥ 0.95, lock is immediate (0 iterations), demonstrating the TSC boost alone lifts coupling efficiency above the sync threshold without any gain adjustment. These results confirm Corollary 6.1 of the formal treatment quantitatively.

6. Empirical Results

The TSC T-U° operator was evaluated within the ASV3-FORMAL self-test suite across 80 benchmark runs spanning 4 defence domains × 4 signal scenarios × 5 state-space dimensions, at 500 governance steps per run.

Table 1: TSC Coherence Index and Stability Metrics (All Scenarios)

Signal Regime	Dim. (d)	TCI Min	TCI Peak	Hyper-Coh %	Anchor Lock	Violations
Nominal	64–1024	0.249	0.892	0%	100%	0
Broadband	64–1024	0.575	1.007	0–61%	100%	0
Adversarial	64–1024	1.033	1.639	32–78%	100%	0
Combined	64–1024	1.029	1.639	31–78%	100%	0

6.1 Dimensional Scaling

The monotone growth of TCI peak with state-space dimension d under adversarial loading (1.033 at d = 64 rising to 1.639 at d = 1024) is among the most significant empirical findings. In higher-dimensional state spaces, AGGL correction vectors are proportionally larger in absolute norm, driving the ultra-kernel norm u★ upward. The √d normalisation in λ(u) prevents unbounded growth without suppressing the scaling trend under active correction.

The practical implication is counterintuitive but important: larger autonomous systems naturally generate stronger coherence signals during active governance. The TSC mechanism is intrinsically better suited to high-capacity architectures, and deploying TSC-aware governance in large autonomous systems yields proportionally larger coupling efficiency gains.

6.2 Universal Anchor Lock and Identity Preservation

Anchor lock (Anch% = 100% across all 80 runs) confirms that the SIF identity functional always classifies the governed state as anchor-locked — the bit entropy of x remains within the information bound at every step. This is stronger than invariant non-violation alone: it confirms

the governance pipeline preserves representational identity at every step, not merely geometric admissibility.

6.3 Hyper-Coherence Prevalence

Under adversarial loading at d = 1024, hyper-coherence is active for 78.4% of governance steps, meaning the system spends the majority of its operating time in the T-U° regime. Under combined signal conditions, prevalence is 77.9%. This confirms that the T-U° state is not a rare edge case but the dominant operating mode under the most demanding governance conditions.

7. Formal Guarantees

The following theorems and propositions constitute the complete mathematical characterisation of the governed hyper-coherence regime.

Thm 3.1	TSC Boundedness: Under ASV3-FORMAL forward invariance, TSC(t) ≤ TSC_max < ∞ for all t ≥ 0.
Prop 3.1	GREEN-Phase Monotone Increase: Under constant sinusoidal input in GREEN phase, $\mathbb{E}$[dTSC/dt] ≥ 0.
Prop 5.1	Bounded Session Duration: Under forward invariance, the RED phase cannot persist indefinitely; any session collapse terminates in at most K steps.
Thm 4.1	Finite Hyper-Coherent Epoch Duration: Every hyper-coherent epoch terminates in at most K_hyper steps under bounded adversarial input.
Cor 6.1	TSC-Assisted Lock Efficiency: N_iters ≤ max(0, ⌈(θ_sync − CE_0) / (g★ · σ · δw)⌉). Higher TSC reduces N_iters; in the hyper-coherent regime, N_iters = 0.

8. Two-Certificate Governance

The TSC operator functions as a real-time representational coherence certificate. This is conceptually distinct from the barrier certificate B(x) ≤ 0, which certifies geometric admissibility. Together they constitute a two-certificate governance space:

Certificate	What It Certifies	Governance Role
`B(x) ≤ 0`	Geometric admissibility (energy-norm containment)	Hard constraint: state within admissible ball
`TSC(t) > 0`	Representational coherence (spectral alignment)	Soft signal: governs coupling tightness via AGSPL

A system presenting high TSC alongside B(x) ≤ 0 is not merely safe — it is operating in a configuration where safety will be maintained with high robustness, because all internal correction mechanisms are pointing in the same direction. A system with B(x) ≤ 0 but low TSC is geometrically safe but representationally fragmented, and may be susceptible to rapid degradation under novel inputs.

9. Limitations and Future Work

Several limitations of the current framework are acknowledged. The hyper-coherence threshold θ_h, boost factor α_TSC, hyper-coherence bonus β_h, and GREEN amplification factor 1.05 are currently set heuristically. A principled calibration method — potentially via Bayesian optimisation or an information-theoretic argument connecting optimal coherence thresholds to spectral admissibility margins — is deferred to future work.

Proposition 3.1 (TSC monotone increase in GREEN) is stated for constant sinusoidal input. Extension to stochastic σ-bounded inputs via a martingale argument for the layer-state norm evolution is a stated objective. All results are currently from software simulation; hardware-in-the-loop validation on a UAV flight controller and a cyber-physical gateway is the next experimental milestone.

The TSC functional treats layer norms additively. A richer model incorporating pairwise cross-layer coherence terms would capture interference effects between governance layers, potentially revealing constructive and destructive resonance phenomena beyond the additive model. Machine-verified proofs of the core theorems in Lean 4 or Z3 are also identified as a priority.

10. Conclusion

The Transient Supra-Coherence (TSC) operator T-U° provides the first formal coherence certificate for layer-state superposition in a real-time governance framework. By aggregating layer-state norms, ultra-kernel projection, and future-adjacency into a single composite index,

TSC drives AGSPL coupling efficiency and activates a discrete hyper-coherence bonus when TSC(t) > θ_h.

Three core guarantees underpin the framework: session collapses are finite (Proposition 5.1), TSC is non-decreasing in expectation during GREEN-phase operation (Proposition 3.1), and hyper-coherent epochs are finite under bounded adversarial input (Theorem 4.1). Empirical results across 80 benchmark runs confirm all theoretical predictions universally, with zero invariant violations, 100% anchor lock, and peak coherence of 1.639 at the highest tested dimension.

Transient Supra-Coherence T-U° establishes that representational alignment and geometric admissibility can be jointly certified at every governance step — and that governed hyper-coherence is not a stochastic accident but a structurally guaranteed emergent property of multi-layer invariant-certified architectures.

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